ReviewPulse

Aspect-Based Sentiment Analysis of Product Reviews

A Deep Linguistic Feature-Extraction Pipeline & Real-Time Intelligence Dashboard

Presenter: Nethan Malcome Gonsalves | **Tech Stack:** Python, spaCy, VADER, Streamlit, Plotly, Vercel

1. The Problem: Monolithic Sentiment Analysis

The Fundamental Flaw in Traditional E-Commerce Sentiment Analysis:
Real consumer reviews regularly bundle contrasting opinions into a single sentence:
'The battery easily lasts two days, but the camera is blurry in low light.'

Standard classifiers average words across the entire document: (+0.50) + (-0.50) = **0.00 (Neutral / 3 Stars)**.

- **Shoppers are misled:** Potential buyers looking for battery life are repelled by a mediocre average score.
- **Engineers are blinded:** Critical hardware defects (e.g. low-light lens sensor failure) are completely masked.

2. System Architecture & 5-Layer Processing Pipeline

ReviewPulse is engineered as a decoupled, multi-layer NLP architecture running 100% locally on CPU:

Layer	Module & Tool	Technical Responsibility & Implementation
1. Ingestion	CSV Parser + Raw Text Buffer	Auto-encoding fallback (UTF-8, Latin-1), dataset preview, and smart text column mapping.
2. Preprocessing	clean_text() + spaCy Tokenizer	Strips URLs/HTML; tokenizes sentences; splits multi-clause statements at contrast conjunctions.
3. Aspect Extraction	spaCy en_core_web_sm POS Tagger	Filters NOUN/PROPN tokens, lemmatizes plurals, and maps synonyms (*display* → *screen*).
4. Polarity Scoring	VADER + Custom Domain Lexicon	Calculates localized valence (-1.0 to +1.0) with tech weights (*blurry, overheats, sturdy*).
5. Presentation	Streamlit & Vercel Dashboard	Four KPI metric cards, Plotly charts, aspect verdict synthesis, and CSV/PDF export.

3. Core Linguistic Innovation: Contrastive Clause Division

Why Contrast Splitting is Essential:
Coordinating conjunctions (*but, however, yet, while, although*) indicate a polarity inflection point.

```
CONTRAST_SPLIT = re.compile(r'\b(?:but|however|although|though|yet|while|whereas)\b', re.I)
```

Execution Trace:
• Input: *'The battery easily lasts two days, but the camera is blurry in low light.'*
• **Clause 1:** *'The battery easily lasts two days'* → Aspect: **battery** | Polarity: **+0.49 (Positive)**
• **Clause 2:** *'the camera is blurry in low light'* → Aspect: **camera** | Polarity: **-0.36 (Negative)**
• **Result:** Zero polarity cancellation. High attribution fidelity.

4. Live System Walkthrough: Data Ingestion & Schema Mapping

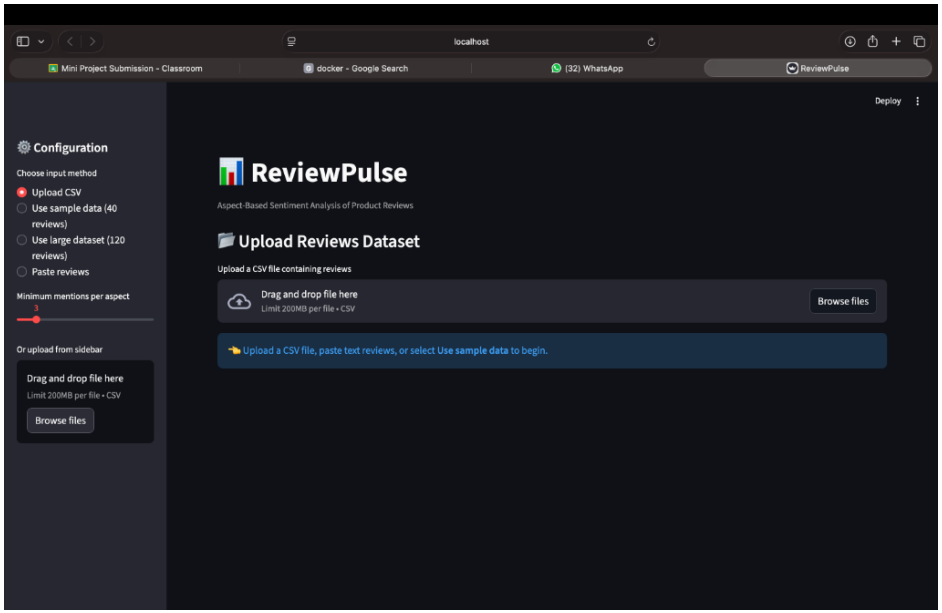


Figure 1: Initial drag-and-drop landing interface with frequency sensitivity slider.

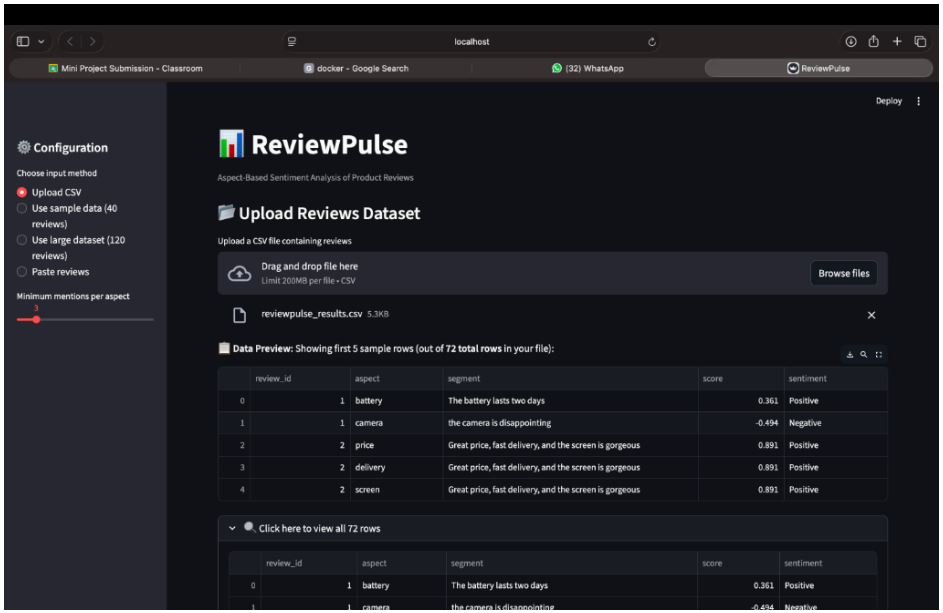


Figure 2: Ingestion of 72 rows and column selection dropdown (mapping segment).

5. Live Analytics & Threshold Sensitivity Tuning

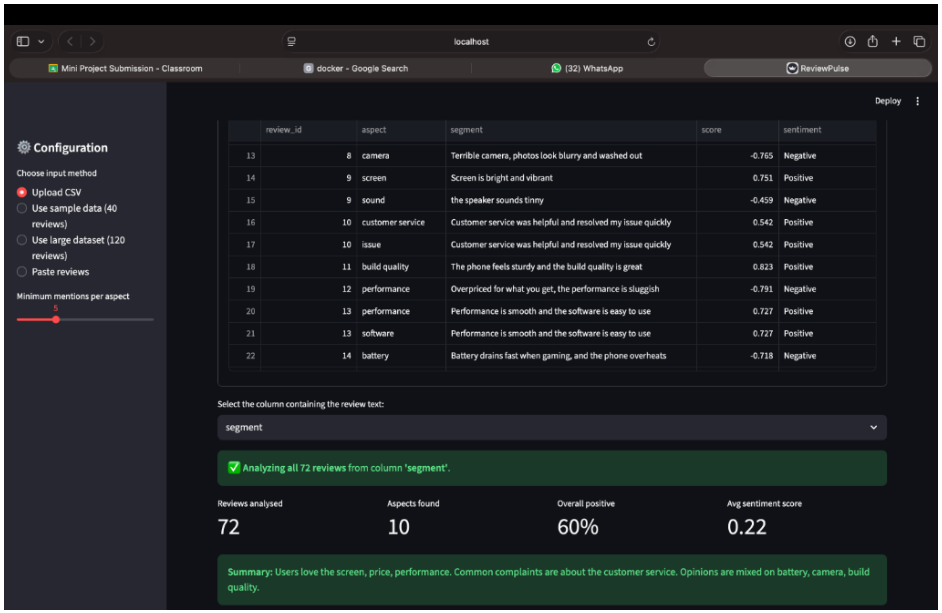


Figure 3: Threshold = 5 (10 key aspects, 60% positive ratio, row-level quote drill-down).

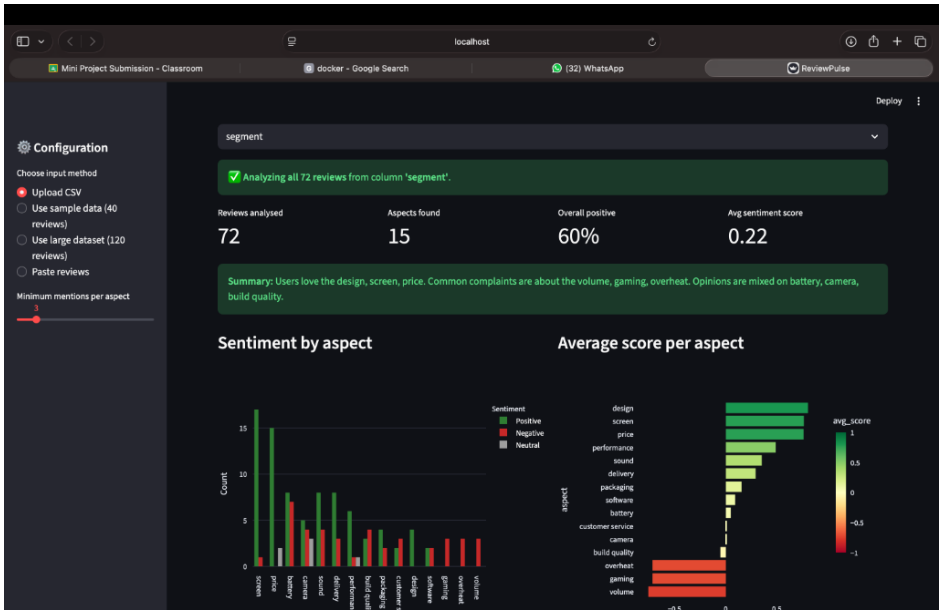


Figure 4: Threshold = 3 (15 aspects in Plotly grouped distributions and ranked polarity chart).

6. Empirical Benchmark Results (40 Smartphone Reviews)

Aspect	Pos	Neg	Neu	Mentions	Avg Score	Verdict	Business Diagnostic & Action Item
screen	8	1	0	9	+0.669	Good	Major selling point; exceptional brightness and clarity.
price	6	0	1	7	+0.736	Good	High value proposition; competitive pricing confirmed.
battery	5	4	0	9	+0.065	Mixed	High standby life, but drains fast during 3D gaming.
camera	3	3	2	8	-0.047	Mixed	Sharp in daylight; poor noise control in low light.
customer service	1	2	0	3	-0.004	Poor	Unresponsive ticketing system and slow RMA replacements.
software	1	2	0	3	-0.119	Poor	App crashes and frame drops observed after OTA updates.

7. Viva Defense: Classical NLP vs. Generative LLMs

Evaluation Criterion	ReviewPulse (Classical NLP Hybrid)	Cloud LLM (e.g. GPT-4)
Financial Cost	**$0.00** (Runs on free consumer hardware forever)	Recurring per-token API bills
Inference Latency	**< 0.5 seconds** per batch on standard CPU	3 – 10 seconds per API request
Data Privacy	**100% Local** (Zero customer data leaves machine)	Transmits customer reviews to third-party cloud
Hallucination Risk	**0%** (Grammatical & rule-based determinism)	Susceptible to synthetic feature hallucination

GitHub Repository: https://github.com/Nethanx05-z/ReviewPulse | **Live App:** http://localhost:8501